

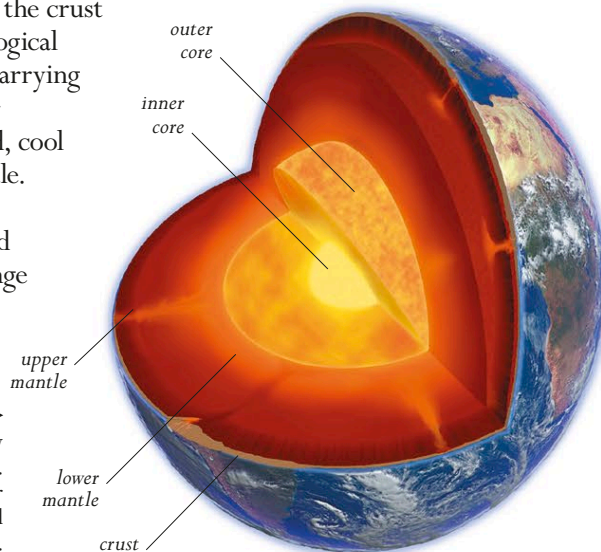
# A LIVING PLANET

Earth is uniquely equipped to support a wide diversity of life, both on land and in the seas. Without heat and light from the Sun, plentiful supplies of water, the protection provided by the atmosphere, and chemicals from the rocks that form the basis of Earth's ecosystems, life would perish.

## DYNAMIC EARTH

Within our Solar System, Earth seems to be uniquely placed to support abundant life. The third planet from the Sun, Earth is neither too close to nor too far from the Sun's heat. It therefore retains an outer atmosphere of oxygen and other gases, and a hydrosphere of plentiful surface water. Together, these form a protective, insulating layer that enables life to flourish. In contrast, the other planets in the Solar System are either too hot or too cold, and devoid of the levels of water and oxygen required to support detectable life.

Earth has a layered structure, with an extremely hot, solid metallic core at its centre, surrounded by an outer molten layer. This, in turn, is surrounded by a thick and hot silicate mantle, which rises to a thin, cool, and brittle outer crust. The mantle is constantly churned by heat rising from the core, which generates new ocean floor rocks and breaks the crust into large "plates". Over geological time, some oceans expand, carrying continents with them. Other oceans become smaller as old, cool crust descends into the mantle. Continents collide forming mountain ranges. Life has had to adapt to the constant change in Earth's environment.



### EARTH'S STRUCTURE >

The liquid mantle is constantly stirred by heat rising from the core. This moves the plates of the outer crust, causing earthquakes and volcanic eruptions on the surface.

## SUN AND MOON

The Sun and the Moon both have a direct impact on life on Earth. Without the Sun's energy, in the form of heat and light, there would be no life. Solar energy heats Earth's atmosphere, the oceans, and the land, producing our varied climate. Because Earth rotates at an angle while it orbits the Sun, the Sun's radiant energy is unevenly distributed over its surface. This results in daily, seasonal, and annual variations in light, heat, and living conditions for plants and animals. Even at the equator, there are marked temperature changes between day and night. The orbit of Earth's satellite moon and its gravitational pull raise tides in Earth's oceans and seas. Tidal cycles are especially influential on coastal life, which has to adapt to changing conditions.

### ^ SOLAR FLARES

The Sun's energy is dramatically released from the surface in periodic explosions, which heat its atmosphere to form solar flares of hot ionized gas.

### WATER AND LIFE

Life depends upon water, which forms over 50 per cent of all living tissues. Most rainfall is driven by evaporation from the oceans, which hold 97 per cent of Earth's surface water, and a vital network of rivers flow in all but the world's hottest, coldest, and driest places.

